

MIG Welding of 316L Stainless Steel

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Alloy and Filler Composition

Note: Minor elements such as S, P, and N, though present in small amounts, significantly affect weldability and corrosion resistance. All C, S, P, and N values are maximum allowed by standards.

Material	C (max)	Cr	Ni	Mo	Mn	Si	S (max)	P (max)	N (max)	Note
316L	0.03 %	16.5– 18.5 %	8– 13%	2– 2.25 %	2%	1.0%	0.04 %	0.001 %	0.11 %	Low carbon, corrosion resistant
ER316L	0.03 %	16– 18%	10– 14%	2–3%	2%	0.75 %	0.045 %	0.03 %	0.10 %	Filler, matches 316L
MIGWELD 316Si	0.025 %	19%	11.5 %	2.75 %	1.75 %	0.7%	0.015 %	0.015 %	-	More Si, good flow
MIGWELD 318NbSi	0.05 %	19%	12%	2.8%	1.8%	0.8%	0.015 %	0.015 %	-	Nb+Si to reduce cracking

Various grades of steel play important roles in the construction of different means of transport.

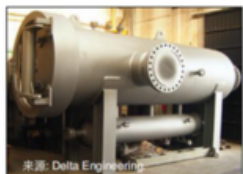
*Sources: Khrais et al., 2022; Szczucka-Lasota & Węgrzyn, 2022
Benato R., F. Dughiero, M. Forzan, A. Paolucci. 2002.*

Applications and Common Problems

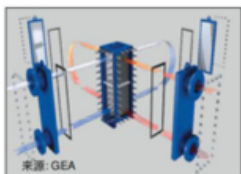
*Sources: [8] Khrais et al., 2022; [9] Szczucka-Lasota & Węgrzyn, 2022; [10] Ghosh et al., 2016
Lee S.J., J.J. Lai. 2003. Klas Weman 2012. Szymański G., A. Patecki. 1984.*

- Applications: Austenitic steels are used in the construction of various means of transport. It is often used in the automotive industry, especially where high corrosion resistance and durability of various
- structural elements are strongly required
- MIG Welding is a spot-welding process using inert shielding gas such as argon (Ar) or a gas mixture of argon and helium (Ar/He) mixture
- Common Issues:
 - - Porosity due to H/O entrapment
 - - Solidification cracking in fully austenitic welds
 - - Intergranular corrosion after welding (The presence of non-metallic inclusions in an austenitic weld provokes cracks because of intercrystalline corrosion)
 - - Hot cracking from excess heat input

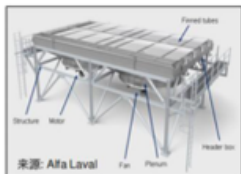
Application



来源: Delta Engineering
分离器
Separators



来源: GEA
板式换热器
Plate Heat Exchanger



来源: Alfa Laval
空气冷却器
Air Coolers



来源: Be Ta
管壳式换热器
Shell & Tube Heat Exchanger



来源: Mech-Tool Engineering
hutaisteel.com



来源: Mech-Tool Engineering
住宿单元
Accommodation Modules

在处理浓缩海水的海水淡化设备中使用的不锈钢，其在含氯离子环境下的点蚀是非常值得注意的。尽管海水冲刷降低了点蚀的趋势，且316L不锈钢是蒸发器的传统材料，但是大多数现代多级闪蒸海水淡化装置(MSF)不使用316L不锈钢。



中兴温德供应的合金316L作为低碳的不锈钢，因其优异的耐腐蚀性在化工行业有着广泛的应用。316L也是属于18-8型奥氏体不锈钢的衍生钢种，



来源: Aker Solutions
钻井隔水管
Drilling Riser



来源: Total
柔性立管
Flexible Risers



来源: Aker Solutions
脐带缆
Umbilicals



来源: Rimor
水下控制单元
Subsea Control Module



来源: Parker
脐带缆终端
Umbilical Termination



来源: KME
控制线
Control Lines

Common MIG Welding Defects

However, the steel tends to crack when welding parameters are poorly selected. Incorrect selection of the process might lead to the formation of various types of non-metallic inclusions such as carbides (for instance $M_{23}C_6$), Cracks in austenitic mobile platform structures might occur when too much heat is introduced during welding.



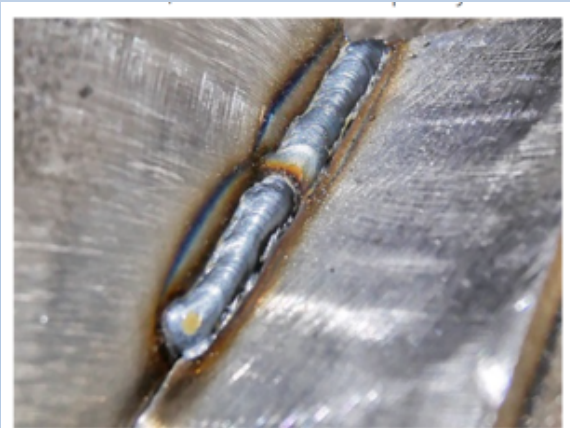
Porosity in welding



Lack of fusion and cold lap



Excessive spatter



Concave and convex weld beads



Burn-through

Skowrońska B., J. Szulc, T. Chmielewski, D. Golański. 2017.

Silva A., B. Szczucka-Lasota, T. Węgrzyn, A. Jurek. 2019.

Mitigation Strategies for Welding Issues

Sources: Khrais et al., 2022; Szczucka-Lasota & Węgrzyn, 2022

Welding Issue	Cause (from paper)	Detailed Mitigation Strategy
Carbide Formation	High heat input promotes M ₂₃ C ₆ carbide precipitation along grain boundaries. This leads to sensitization and intergranular corrosion.	- Use low-carbon filler metal (e.g., ER316L)- Keep interpass temperature < 110°C- Avoid prolonged high temp near melting point
Porosity	Caused by excessive shielding gas flow or contaminated surfaces trapping gas bubbles in the weld.	- Clean base metal before welding- Use proper gas flow rate (15–20 L/min)- Maintain arc stability with correct voltage
Lack of Fusion	Low heat input (low current or high travel speed) fails to fully melt base material.	- Increase welding current (e.g., from 100A to 112A)- Slow down travel speed slightly- Maintain proper torch angle (~15°)
Incomplete Penetration	Inadequate energy due to low current or rapid movement of torch	- Use optimal current (e.g., 112A)- Avoid travel speeds > 25 mm/s- Apply weaving technique for wide joints
Grain Coarsening	Excessive heat input (e.g., high current like 124A) increases grain size, reduces mechanical strength.	- Avoid using current > 120A- Use pulsed MIG welding to reduce heat input- Allow natural cooling between passes
Hot Cracking	Fully austenitic microstructure cannot absorb impurities during solidification, causing cracks.	- Use filler metals that produce slight δ-ferrite (~3–7%)- Control cooling rate to avoid fast solidification- Use balanced Cr/Ni ratio

Thermo-Calc Simulation Draft – MIG Welding of 316L Stainless Steel

Simulation Setup

- Database: TCFE13
- Method: Scheil–Gulliver non-equilibrium solidification
- Alloy composition: Fe 69.22 wt%, Cr 16 wt%, Ni 10 wt%, Mo 2 wt%, Mn 2 wt%, Si 0.75 wt%, C 0.03 wt% (matches 316L MIG filler)
- Cooling rate: 50 °C/s (typical for MIG weld pool cooling)
- Justification:
 - Scheil captures rapid, local solidification in welds
 - 50 °C/s approximates measured HAZ cooling rates in 316L MIG welding

Phase Coexistence and Solidification Range

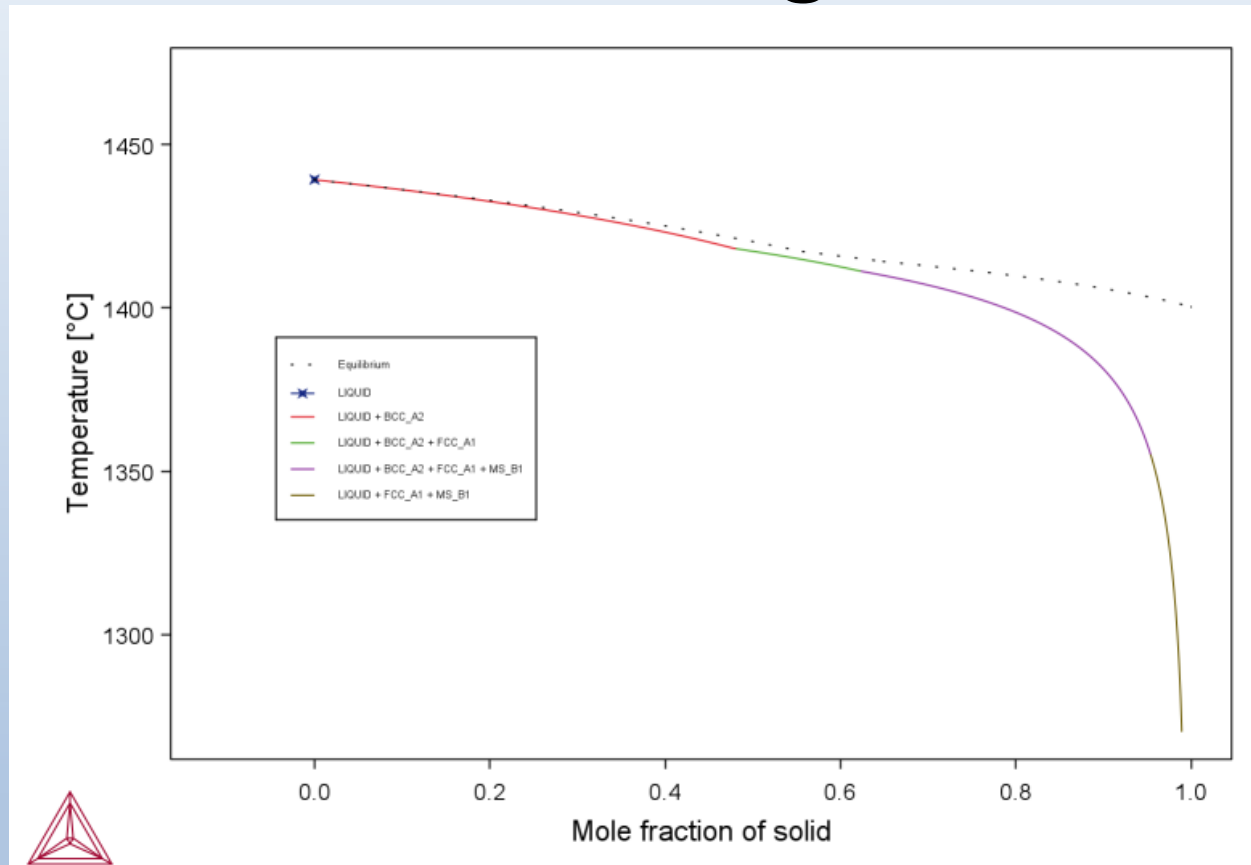
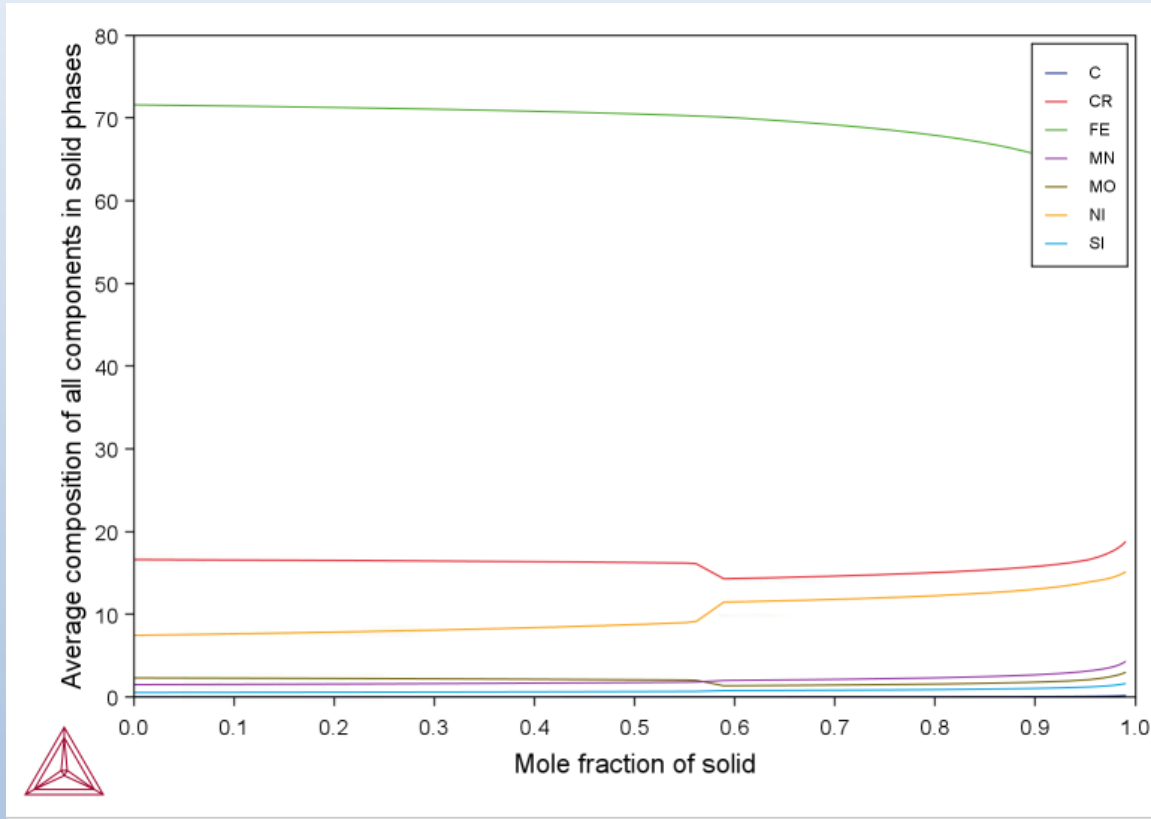


Figure: cheil simulation showing non-equilibrium solidification path for 316L stainless steel highlighting the formation of BCC_A2, FCC_A1, and MS_B1 phases over increasing solid fraction.

Average Composition in Solid Phases During Cooling



Mo and Cr decrease in concentration near solidus → potential segregation at grain boundaries

Interpretation and Key Observations

Solidification begins with dual-phase coexistence (LIQUID + BCC_A2), later transitioning to FCC_A1 as the dominant phase.

A wide solidification range (~165 °C from 1442 °C to 1277 °C) indicates potential for solidification cracking.

Cr and Mo exhibit segregation at grain boundaries, which may lead to brittle phase formation or intergranular corrosion.

Elemental segregation may impact post-weld corrosion resistance and mechanical properties.

Thermo-Calc Simulation Summary

1. Simulation Method and Context:

- Scheil-Gulliver model selected to simulate rapid solidification during MIG welding.
- Mimics non-equilibrium conditions in weld zones.

2. Elemental Partitioning Observations:

- Cr and Mo show delayed solidification and remain in the liquid phase longer → grain boundary enrichment.
- Risk of intergranular corrosion or formation of secondary brittle phases.

3. Solidification Temperature Range:

- Liquidus $\approx 1442\text{ }^{\circ}\text{C}$, Solidus $\approx 1277\text{ }^{\circ}\text{C}$ → Range $\approx 165\text{ }^{\circ}\text{C}$.
- Such a large temperature span increases likelihood of hot cracking.

4. Conclusion:

- Solidification path involves multiple phase transformations (BCC_A2 → FCC_A1 → MS_B1).
- Careful heat input control is essential to limit segregation and cracking risks.

Detailed Elemental Partitioning Behavior

- - Cr (Chromium): Decreases sharply between 1460–1430°C, then slightly recovers. Indicates early segregation to liquid and partial late reabsorption. High risk of grain boundary enrichment and intergranular corrosion.
 - Mo (Molybdenum): Slight increase in solid phase near 1435°C. Suggests it remains in liquid longer and segregates toward grain boundaries, potentially forming brittle phases.
 - Ni (Nickel): Steadily increases during cooling, with a sharper rise near 1435°C. Indicates delayed solidification and good solubility. Low segregation tendency.
 - Mn (Manganese): Minor decrease overall, relatively stable in solid phase. Moderate segregation behavior, unlikely to pose welding risks.
 - Si (Silicon): Virtually unchanged throughout cooling. Strong solubility in solid phase and minimal segregation risk.
 - C (Carbon): Extremely low content with minimal visible variation. Not prominent in this simulation but may form carbides under certain conditions.
 - Fe (Iron): Shows consistent increase due to relative enrichment as other alloying elements segregate. Becomes dominant in final solid phase.

References

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